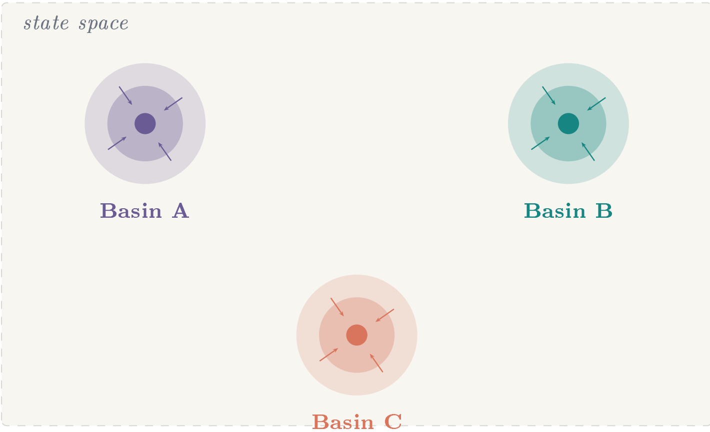
Sparse Koopman Autoencoders for Multibasin Dynamical Systems

Induced latent sparsity makes the active support a label-free indicator of the relevant basin and local linear law.

01 • The Problem

Multibasin systems break a single global linearization

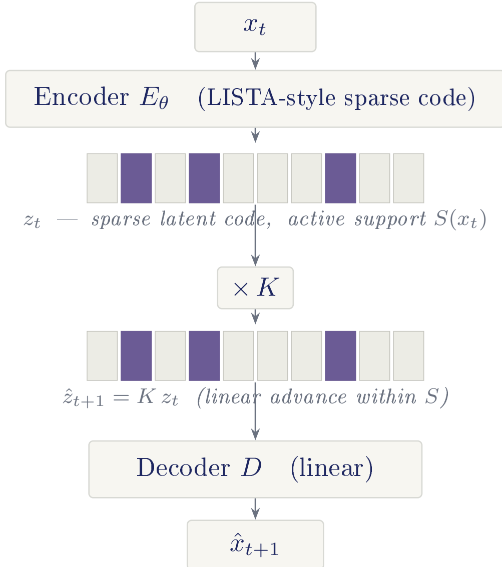


A single global linear K cannot describe all basins simultaneously.

Each basin admits a local linear description, but they are mutually incompatible.

02 • The Model

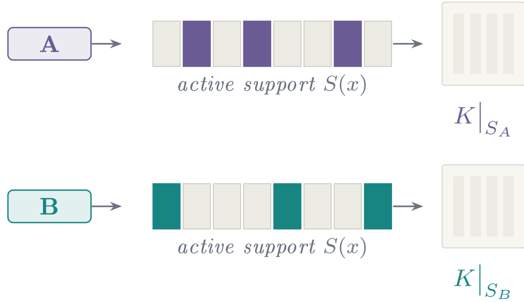
Sparse-coding Koopman autoencoder



Periodic re-encoding. Decode and re-encode every m steps so the support can change as the predicted state moves.

03 • The Key Idea

The active support tells us the basin and the local linear law



Different basin $\rightarrow$ **different active support** $\rightarrow$ **different active columns of** K $\rightarrow$ different effective local linear law.

Periodic re-encoding refreshes the support across regimes:

z_A → K^m → decode → encode → z_B

→ *the support can switch when the trajectory crosses into another basin.*

Core idea. **Sparse latent code** $\rightarrow$ **active support agrees with basin** $\rightarrow$ **support routes prediction to a local linear law** *(no basin labels used during training).*